

NOTES FOR APSC 2113

ENGINEERING ANALYSIS - I

PROF. KARTIK BULUSU

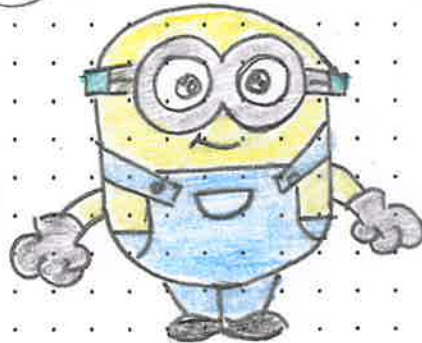
DEPARTMENT OF MECHANICAL AND
AEROSPACE ENGINEERING

THE GEORGE WASHINGTON UNIVERSITY



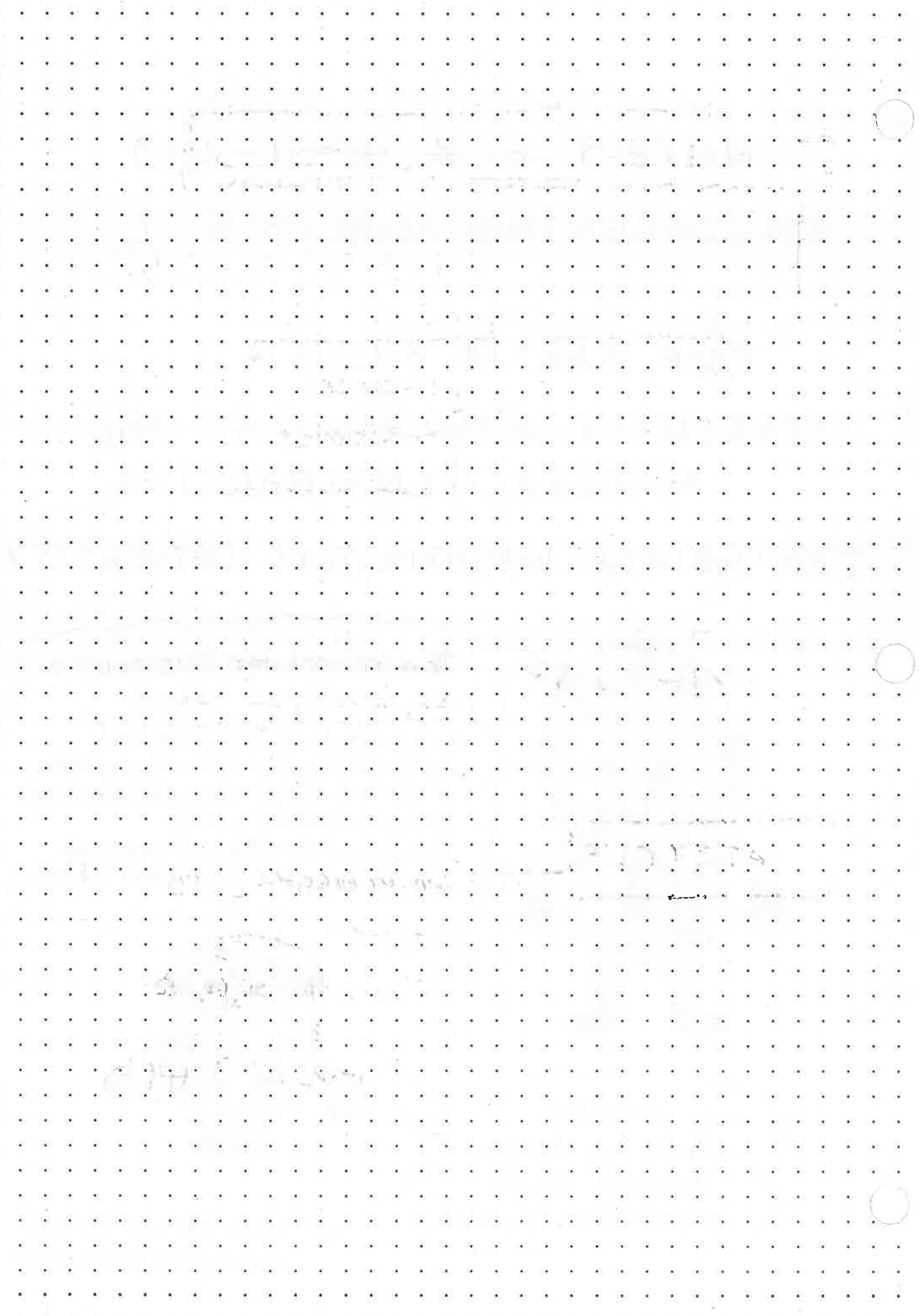
The notes for this course were developed using

1. Elementary Differential Equations and Boundary Value Problems by C. Henry Edwards & David E. Penney, and
2. Differential Equations and Linear Algebra by Gilbert Strang

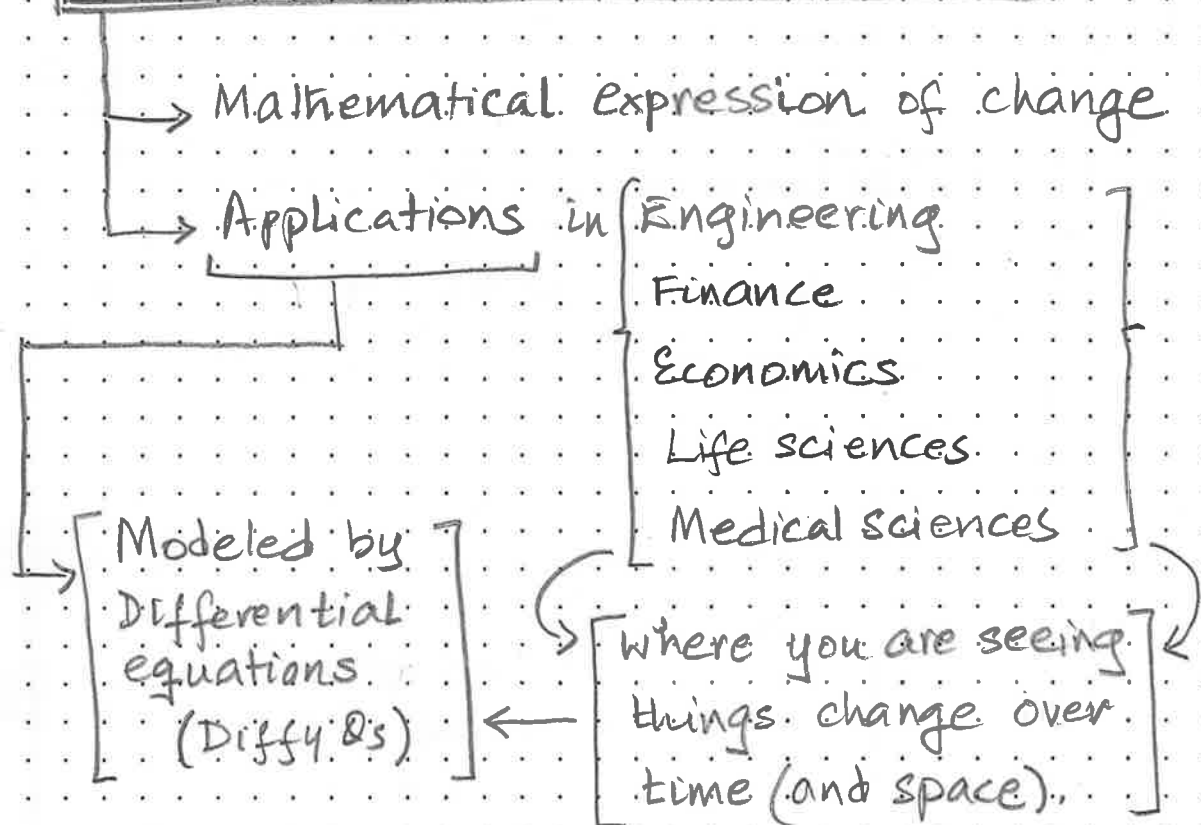


And, youtube videos by Arthur Mattuck and Gilbert Strang, MIT.

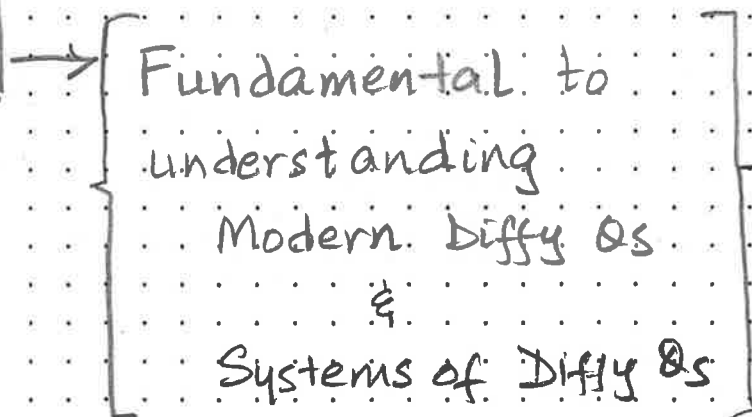
© 2026 Kartik Bulusu.
All rights reserved.



DIFFERENTIAL EQUATIONS



MATRICES



IDEAS OF THE FIRST LECTURE ON ORDINARY DIFFERENTIAL EQUATIONS (ODEs)

Defⁿ:

DIFFERENTIAL EQUATION:

An equation relating an unknown function and one or more of its derivatives.

$$\frac{dx}{dt} = f'(t)$$

function, f
is the rate at
which $x = f(t)$
is changing with
respect to the
independent variable,
 t .

Laws of the universe are expressed
in mathematics.

ALGEBRA

- To solve static problems that don't involve change
- We seek unknown numbers.

DERIVATIVES

- To solve problems involving change.
- We seek unknown functions.

WHAT IS REASONABLE TO LEARN ABOUT ODEs?

[ODE: Derivative of y i.e., $\frac{dy}{dt}$, depends on y .]

FIRST-ORDER ODEs:

$$\frac{dy}{dt} = f(t, y)$$

[General form]

→ First-derivatives come into the equations.

→ Rate of change of the unknown ($\frac{dy}{dt}$) partly dependent on,

- ① the solution (y) itself, and
- ② inputs, $q(t)$ that produce their own change

→ constant
Linear: $f(y) = a y$ OR $a(t)$

Non-Linear: $f(y) = a g(y)$

Constant or $a(t)$ or $a(y)$

$y^2, y^3, \dots$, sines, cosines, exponents or combinations thereof.

$$\frac{dy}{dt} = a y + q(t)$$

Linear: ① We see y by itself!

② $a \rightarrow$ constant or $a(t)$

SECOND-ORDER ODES:

$$m y'' + b y' + k y = q(t)$$

→ Second-derivatives come into the equations

$$\frac{d^2 y}{dt^2} = y'' = -k y$$

Linear:

Acceleration on left-hand-side equals some force, that depends linearly on y .

Newton's Law force:

Acceleration times constant mass (m).

Damping force:

Friction slowing it down by the first-derivative (velocity)

Another term:

Depending on y , itself

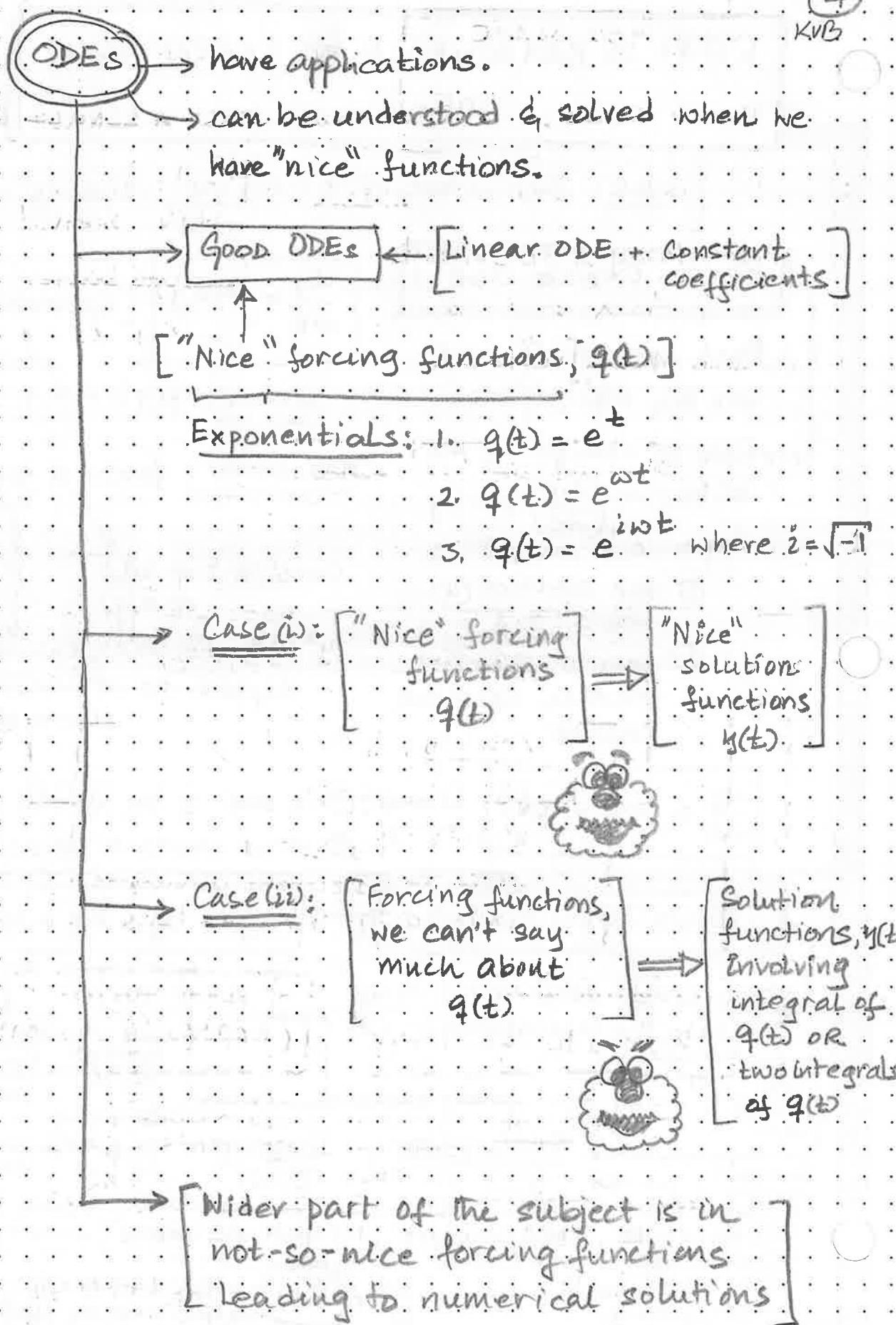
Outside force:

Some person or machine creating movement

$q(t)$

→ FORCING TERM:

Goes into the system & becomes a part of y .
Grows, decays, oscillates with y .



WHEN YOU HAVE

MANY ODES,

NOT, JUST A SINGLE ODE.

- Many springs with masses
- Many bank deposits
- Many bacteria
- Many resistors

- One spring with mass
- One bank deposit
- One bacterium
- One resistor

SYSTEM OF "n" ODES

$$\frac{dy}{dt} = Ay$$

First-order
ODEs with
constant
coefficients

Second-order
ODEs with
constant
coefficient

$$\frac{d^2y}{dt^2} = -Sy$$

Solution: $y(t)$

→ $y(t)$ is a vector

$$y(t) = [y_1, y_2, y_3, \dots]$$

as in "n" different unknowns
in "n" different equations

A: $n \times n$ matrix
(hopefully symmetric)

S: $n \times n$ matrix
(hopefully symmetric)

Solve: Several Linear ODEs
+ constant coefficients
at once

KEY IDEAS: ① EIGENVALUES &
② EIGEN VECTORS

← This key bit of
linear algebra
uncouples the "n"
equations to solve them
separately.

THE CALCULUS WE NEED TO KNOW.
IT'S NON-NEGOTIABLE!

6.
KVIB

BASIC DERIVATIVES

y	x^n	$\sin x$ $\cos x$	e^x	$\ln x$
$\frac{dy}{dx}$	$\frac{x^{n+1}}{n+1}$	$\cos x$ $-\sin x$	e^x	$\frac{1}{x}$

RULES

y	$\frac{dy}{dx}$	
$f+g$	$(df + dg)$ OR $(f' + g')$	LINEAR OPERATION
$f(x)g(x)$	$f'g + fg'$	PRODUCT
$\frac{f(x)}{g(x)}$	$\frac{gf' - fg'}{g^2}$	QUOTIENT
$f(g(x))$	$f'(g(x))g'(x)$ OR $\frac{df}{dg} \cdot \frac{dg}{dx}$	CHAIN RULE

FUNDAMENTAL THEOREM OF CALCULUS

Derivative of the integral of a function, $y(x)$, is the function itself.

$$\frac{d}{dx} \int_0^x y(p) dp = y(x)$$

limit of integration

dummy variable

EXAMPLE THAT DEMONSTRATES THE FUNDAMENTAL THEOREM OF CALCULUS

$$y(t) = \int_0^t e^{t-s} q(s) ds \text{ solves the ODE } \left[\frac{dy}{dt} = y + q(t) \right]$$

Note: We can derive this formula this ODE

We are going to check the fundamental calculus theorem is a solution to the ODE.

START: $y(t) = \int_0^t e^{t-s} q(s) ds$

$$\Rightarrow y(t) = \underbrace{e^t}_{f(t)} \underbrace{\int_0^t e^{-s} q(s) ds}_{g(t)} \leftarrow \left\{ \begin{array}{l} \text{I can take the derivative} \\ \text{and use the product rule} \end{array} \right\}$$

$$\begin{aligned} \Rightarrow \frac{dy}{dt} &= \frac{d}{dt} \left[e^t \int_0^t e^{-s} q(s) ds \right] \\ &= \frac{d}{dt} e^t \left[\int_0^t e^{-s} q(s) ds \right] + e^t \frac{d}{dt} \left(\int_0^t e^{-s} q(s) ds \right) \\ &= e^t \int_0^t e^{-s} q(s) ds + \cancel{e^t} \cancel{e^{-t}} q(t) \\ &= \underbrace{\int_0^t e^{t-s} q(s) ds}_y + q(t) \Rightarrow \boxed{\frac{dy}{dt} = y + q(t)} \end{aligned}$$

Use the fundamental calculus theorem

END: This is where we started!

TANGENT LINE TO A GRAPH

↳ Suppose we have any function, $f(t)$

↳ We know something about the function at one point

↳ We want to know the function at a point little beyond, t . i.e., Δt

↳ $f(t + \Delta t)$

$$f(t + \Delta t) \approx f(t) + \Delta t \cdot \frac{df(t)}{dt}$$

{ function at point, t } { correction term }

first-derivative at point, t
OR Slope

$$\Rightarrow f(t + \Delta t) - f(t) \approx \Delta t \cdot \frac{df(t)}{dt}$$

$$\Rightarrow \frac{\Delta f}{\Delta t} \approx \frac{df(t)}{dt}$$

Think of this as Δf

Classic idea of calculus!

The derivative of $f(t)$ at point, t is quite close to $\frac{\Delta f}{\Delta t}$, i.e., for changes over a very short time interval (Δt).

let's look at $f(t + \Delta t)$ again!

$$f(t + \Delta t) \approx f(t) + \Delta t \cdot \frac{df(t)}{dt}$$

This is the tangent line

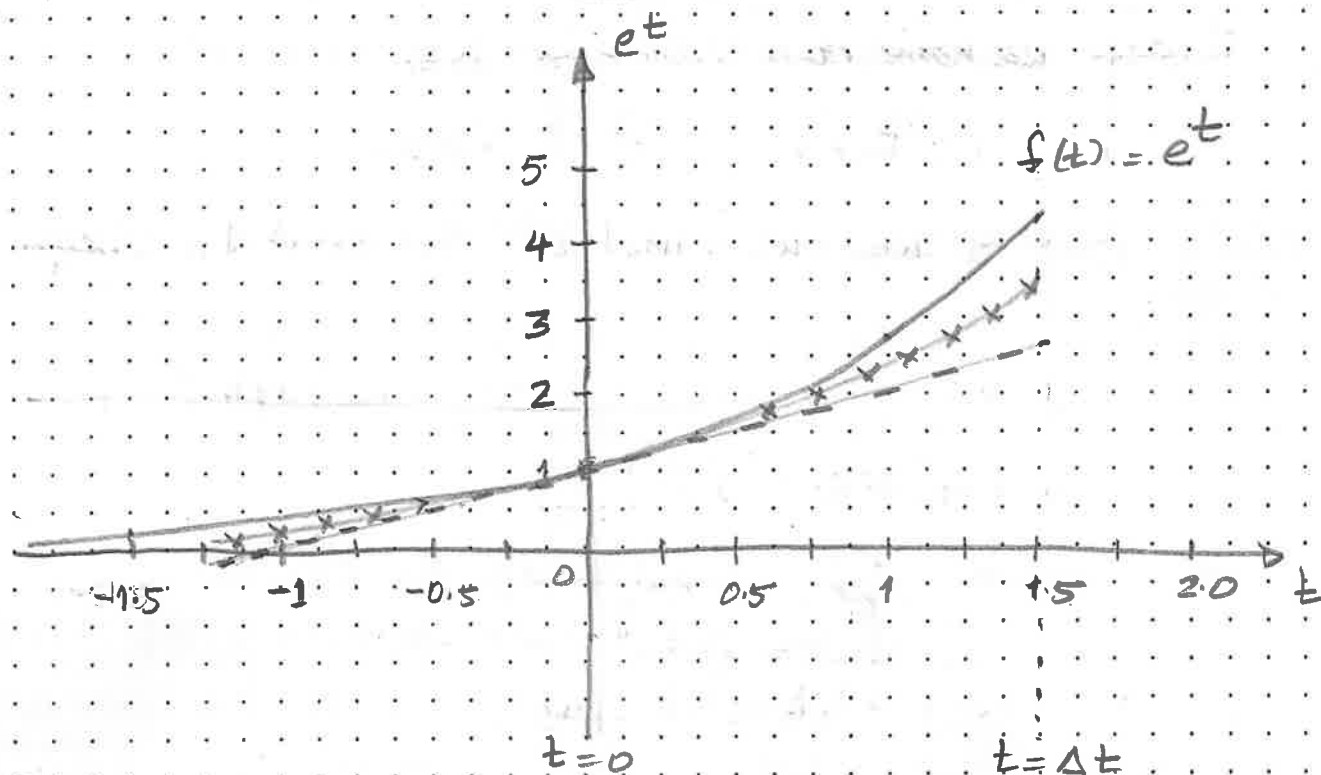
constant term

It's a function of Δt Slope

↳ This is similar to $y = c + x \cdot m$

EXAMPLE GRAPH OF e^t

9
KVB



Scenario: If Δt is large, the line is far from the actual curve. (See line: - - -)

④ We want to get the - - - line closer to the actual curve

⑤ How can we do that? By bending the line?

⑥ How can we bend the line? Take the 2nd derivative

• We put in an extra term in our equation for $f(t + \Delta t)$

$$\Rightarrow f(t + \Delta t) \approx f(t) + \underbrace{\Delta t \frac{df(t)}{dt}}_{\substack{\text{1st derivative} \\ \text{or slope}}} + \underbrace{\frac{1}{2} (\Delta t)^2 \frac{d^2 f(t)}{dt^2}}_{\substack{\text{2nd derivative} \\ \text{Bends the tangent} \\ \text{2nd correction term}}}$$

1st correction term

Tangent line to the curve

changes the tangent line to tangent parabola. (See line: -x-x-)

→ What if we want to "bend it better,"

→ or get the new line to do a better job matching $f(t)$?

→ We put in the 3rd derivative, $\frac{d^3 f(t)}{dt^3}$
and the 4th derivative, $\frac{d^4 f(t)}{dt^4}$,
and so on...

→ If we put in "all" the derivatives, we get the line to match the function, $f(t)$; we get the function $f(t)$.



In general,

We can recreate a function, $f(t)$, by knowing at some point, t ,

- ① It's height $[f(t)]$,
- ② It's slope $[\frac{df}{dt}]$,
- ③ It's bending $[\frac{d^2 f}{dt^2}]$,
- ④ and all other terms

TAYLOR SERIES

$$f(t + \Delta t) = \underbrace{f(t)}_{\text{height}} + \underbrace{\Delta t \frac{df}{dt}}_{\text{slope}} + \frac{1}{2!} (\Delta t)^2 \frac{d^2 f}{dt^2} + \dots + \frac{1}{n!} (\Delta t)^n \frac{d^n f}{dt^n} + \dots$$

height slope bending

After thoughts about Taylor series.

① It's frightening; it's got infinitely many terms!

② The tangent & parabola are practical.



③ Higher-order derivative terms are less practical.

④ A counter-example to point #3: $f(t) = e^t$

$$\text{For } f(t) = e^t, \quad \frac{d e^t}{dt} = \frac{d^2 e^t}{dt^2} = \dots = \frac{d^n e^t}{dt^n} = e^t$$

INTEGRALS AS GENERAL & PARTICULAR SOLUTIONS OF FIRST-ORDER ODES

$$y' = f(x)$$

↳ independent variable

$$\frac{dy}{dx} = f(x)$$

① Linear, separable 1st order ODE

STEP 1: Integrate equation ① on both sides

GENERAL
SOLUTION

$$\int dy = \int f(x) dx + C$$

$$\Rightarrow y(x) = \int f(x) dx + C$$

Arbitrary
constant of
integration

② General solution
of equation ①,
because it involves
an arbitrary
integration constant

Suppose we found some, $G(x)$ such that

$$G'(x) \equiv f(x),$$

then

$$y(x) = G(x) + C$$

Meaning:
We could integrate
 $f(x)$ and find
the output of it

③ Another form of
the general
solution
that can take in a range
of values: $C \in [C_1, C_2, C_3, \dots]$

$y(x) = G(x) + C$ is the general solution where $C = [C_1, C_2, C_3, \dots]$

$$y_1(x) = G(x) + C_1$$

$$y_2(x) = G(x) + C_2$$

$$y_3(x) = G(x) + C_3$$

These are all valid solutions; each for a unique or particular value of C .

EXAMPLE

Suppose we have a linear, separable, ODE $\frac{dy}{dx} = \frac{x}{2}$ and we want to solve this ODE

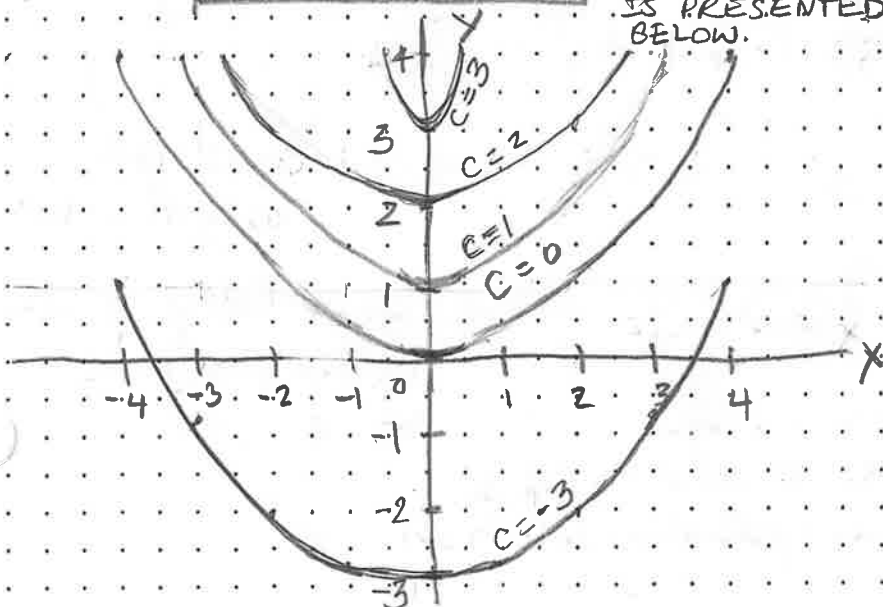
General solution: $dy = \frac{1}{2} x dx$

Integrating on both sides

$$\int dy = \frac{1}{2} \int x dx + C$$

$$\Rightarrow y(x) = \frac{x^2}{4} + C$$

GRAPHICALLY, THIS SOLUTION IS PRESENTED BELOW.



- We have a family of solutions.
- Each solution is plotted within the same domain.
- Notice that each C_n
 - is geometrically the vertical distance
 - between each solution
 - in a "parallel" sense.

Particular solution:

(14) KVB

Recall: General solution of Equation ①

$$\left[\frac{dy}{dx} = f(x) \right]$$

is equation ③ $[y(x) = G(x) + C]$

→ As seen in the previous example, we have a family of solutions, generated from a general solution.

For obtaining a particular solution of equation ①,

We need:

④ A particular value of C

We can get the particular value of C

by satisfying an initial condition

START: $\frac{dy}{dx} = f(x) \rightarrow$ ① $[y(x_0) = y_0]$
Given ODE

$y(x) = G(x) + C \rightarrow$ ③ Recall the general solution

$y(x_0) = y_0 \rightarrow$ ④ Generally, given in the problem statement

Substitute the initial condition, eq ④ in the general solution, eq ③

$$\Rightarrow y(x_0) = G(x_0) + C$$

$$\Rightarrow y_0 = G(x_0) + C \longrightarrow \textcircled{5} \text{ From initial condition equation}$$

$$\Rightarrow \boxed{C = y_0 - G(x_0)}$$

$\textcircled{6}$ Value of C ,
calculated from
the initial condition.

The particular solution is
now obtained by
substituting $\textcircled{6}$ into the
general solution $\textcircled{3}$:

$$\boxed{y(x) = G(x) + [y_0 - G(x_0)]}$$

SUMMARY:

$\textcircled{I}$ GENERAL SOLUTION $\longrightarrow$ involves an
arbitrary integration
constant, C

$\textcircled{II}$ PARTICULAR SOLUTION $\longrightarrow$ involves



appropriate
choice of C

Satisfying
some initial
condition

$$y(x_0) = y_0$$

EXAMPLE ①:

$$\frac{dy}{dx} = 2x + 3$$

Solve this
ODE for the
initial condition

$$y(1) = 2$$

SOLUTION:

STEP ①

$$\frac{dy}{dx} = 2x + 3$$

Find general solution

$$\Rightarrow dy = (2x + 3) dx$$

Integrating on both sides

$$\Rightarrow \int dy = \int (2x + 3) dx + C$$

arbitrary
integration
constant

$$\Rightarrow y(x) = \int 2x dx + \int 3 dx + C$$

$$= \frac{2x^2}{2} + 3x + C$$

$$y(x) = x^2 + 3x + C$$

General
solution

STEP ②

Satisfy initial condition to find the particular solution

$$y(1) = 2 \rightarrow \text{Given initial condition } [y(x_0) = y_0]$$

→ Substitute in the general solution from step ①

$$y(1) = 1^2 + 3(1) + C = 2$$

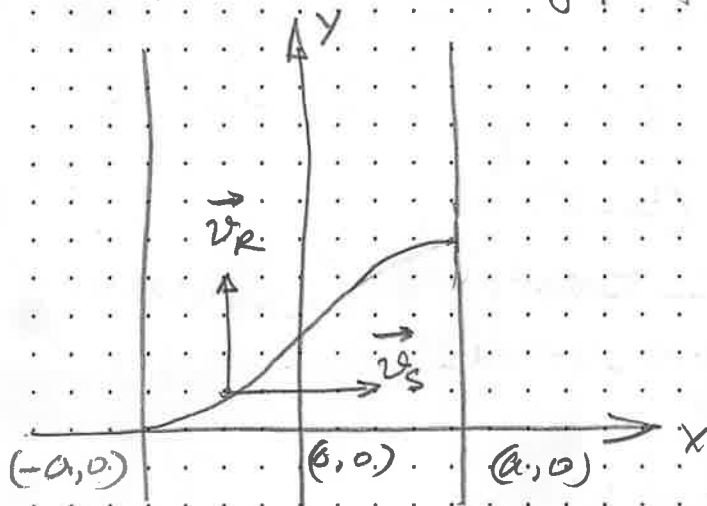
$$\Rightarrow 4 + C = 2 \Rightarrow C = -2$$

→ Substitute the value of C in the general solution

$$\Rightarrow y(x) = x^2 + 3x - 2$$

✓

EXAMPLE (2): A swimmer is attempting to swim across a river at a velocity, v_s . v_R is the velocity of the water flow. The banks of the river are located at $x = -a$ and $x = a$, along with the velocity profile as shown in the figure.



The velocity profile equation is given as

$$v_R = v_0 \left(1 - \frac{x^2}{a^2} \right)$$

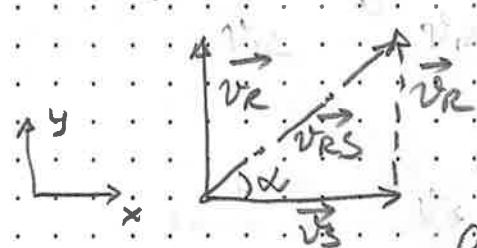
where $v_R = v_0$ at center ($x = 0$)

and

$v_R = 0$ at the banks $x = -a$ & a .

- ① Find the governing ODE of the swimmer problem.
- ② Find the general & particular solutions for the ODE describing the swimmer's trajectory and drift for the following conditions:
The river is 1 mile wide, $v_0 = 9$ miles/h and $v_s = 3$ miles/h; find how many miles does the swimmer drift.

Solution ①: Let's construct the vector diagram of velocities



$$\tan \alpha = \frac{|v_R|}{|v_s|} = \frac{v_R}{v_s}$$

$$\tan \alpha = \frac{dy}{dx} = \text{slope} \quad \therefore \frac{dy}{dx} = \frac{v_R}{v_s}$$

Recall: $v_R = v_0 \left(1 - \frac{x^2}{a^2} \right)$

$$\frac{dy}{dx} = \frac{v_0}{v_s} \left(1 - \frac{x^2}{a^2} \right)$$

→ This is the governing ODE, where the solution $y(x)$ represents the swimmer's trajectory.

Solution (2)!Given: $v_0 = 9$ miles/h $v_s = 3$ miles/h

River width (2a) = 1 mile

$$\Rightarrow a = \frac{1}{2} \text{ miles}$$

Substituting the above values in equation (1)

$$\frac{dy}{dx} = \frac{9}{3} \left(1 - \frac{x^2}{(\frac{1}{2})^2} \right)$$

$$\Rightarrow \boxed{\frac{dy}{dx} = 3(1 - 4x^2)} \rightarrow (2)$$

Separating the variables

$$dy = 3(1 - 4x^2) dx$$

Integrating on both sides

$$\int dy = \int 3(1 - 4x^2) dx + C \quad \text{Arbitrary constant}$$

$$\Rightarrow y(x) = \int 3 dx - \int (3)(4)x^2 dx + C$$

$$\Rightarrow y(x) = 3x - \frac{4x^3}{\cancel{3}} + C$$

$$\Rightarrow \boxed{y(x) = 3x - 4x^3 + C} \rightarrow \text{General Solution} \rightarrow (3)$$

Equation (3) is the general solution that describes the swimmer's trajectory.

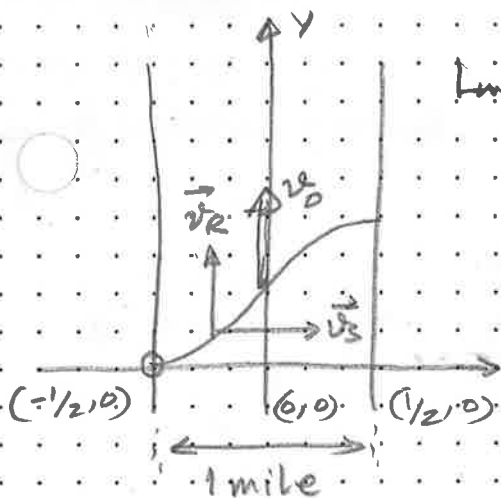
Note: In this problem we were not provided an initial condition, explicitly,

hms. We, therefore, should find an appropriate condition to apply and find the value of the arbitrary constant.

hms. Once, we find the appropriate value of C , we can come up with the particular solution.

To find an appropriate initial condition:

hms. Let's look at the figure again.



hms. From the figure we understand that the swimmer is at the starting point of the swim at $(-1/2, 0)$.

hms. Therefore, there is no drift miles at this point.

$$\Rightarrow y(x) = 0 \text{ at } x = -1/2$$

$$\Rightarrow \boxed{y(-1/2) = 0} \rightarrow \text{Initial condition!}$$

Substituting the initial condition

into the general solution, equation (4),

$$y(-1/2) = 3(-1/2) - 4(-1/2)^3 + C = 0 \quad (5)$$

$$= -\frac{3}{2} + \frac{4}{8} + C = 0 \Rightarrow \boxed{C = 1} \quad (6)$$

∴ Particular solution:

$$\boxed{y(x) = 3x - 4x^3 + 1}$$

Describes the swimmer's trajectory.

Finally, we can calculate the distance in miles the swimmer drifted while swimming across a 1 mile wide river.

- The river is 1 mile wide.
- The swimmer swims from point $(-\frac{1}{2}, 0)$ to the point $(\frac{1}{2}, 0)$.
- We need to know the drift at point $(\frac{1}{2}, 0)$.

$y(\frac{1}{2}) = 3(\frac{1}{2}) - 4(\frac{1}{2})^2 + 1 \rightarrow$ Using the particular solution

$y(\frac{1}{2}) = \frac{3}{2} - \frac{1}{2} + 1$

$\Rightarrow y(\frac{1}{2}) = 2 \text{ miles} \rightarrow$ Swimmer drift

METHODS TO SOLVE ODEs

ANALYTICAL

$$y' = f(t, y)$$

$y_1(t)$: Particular Solution

GEOMETRIC

Direction field

Integral curve

NUMERICAL

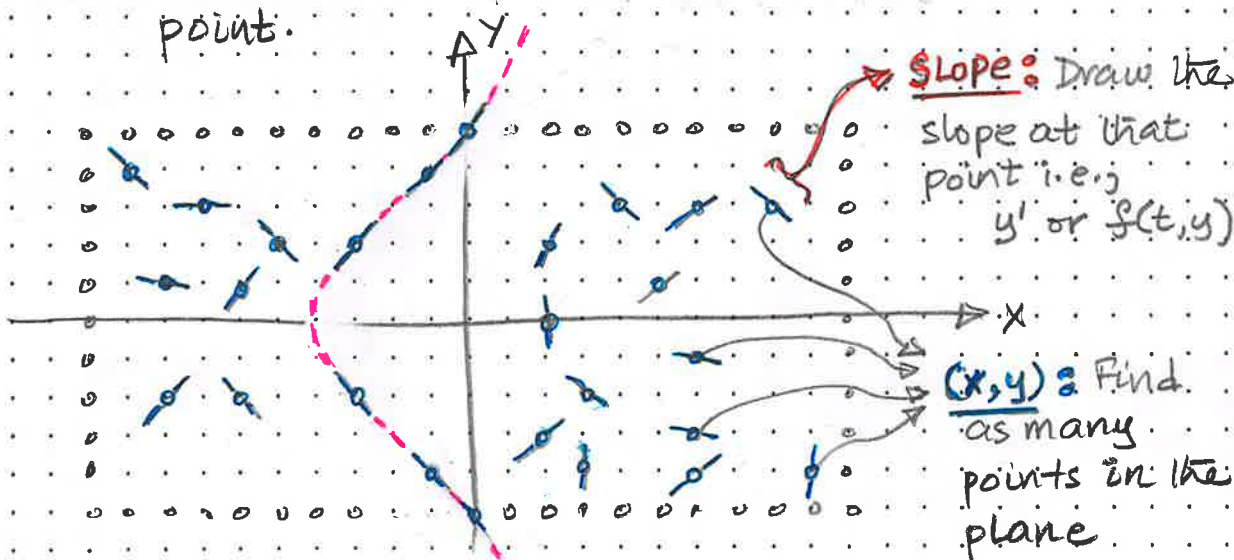
PROGRAMMING

MATLAB: ode45()

PYTHON: numpy, scipy

What is a direction field?

Plane of infinite points representing the ODE with the direction (or slope), y' , at each point.



INTEGRAL CURVE:

HAS THE DIRECTION OF THE FIELD AT ALL POINTS ON THE PLANE

TANGENT LINE JOINING THE SLOPES

REPRESENTS A PARTICULAR SOLUTION TO THE ODE

CURVES CANNOT INTERSECT OR CROSS

SUMMARY:WRITING DOWN THE SOLUTION OF THE ODEANALYTICALLY $\Rightarrow$ IS THE SAME AS $\Leftarrow$ DRAWING THE DIRECTION FIELD & INTEGRAL CURVEGEOMETRICALLY

AND

VICE VERSA

PROOF

$$\left\{ \begin{array}{l} y_1(x) \text{ is a} \\ \text{solution to the} \\ \text{ODE: } y' = f(x, y) \end{array} \right\}$$


$$\left\{ \begin{array}{l} \text{Graph of } y_1(x) \\ \text{is an integral} \\ \text{curve in the} \\ \text{direction field of} \\ y' = f(x, y) \end{array} \right\}$$
At a point $[x, y_1(x)]$

Plugging $y_1(x)$ into
the equation of ODE;
it must satisfy

$$\Rightarrow y_1'(x) = f(x, y_1(x))$$

$$\left(\text{slope of } y_1(x) \right) = \left(\text{slope of the} \right. \\ \left. \text{direction field } f(x, y_1(x)) \right)$$

Right-hand-side in both equations
are the same

∴ Slopes in both equations
should be the same.

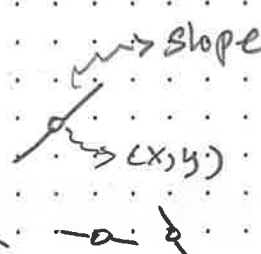


How do you go about solving $y' = f(x, y)$ geometrically?

By computer:

1. Pick points (x, y) on (equally spaced) a plane

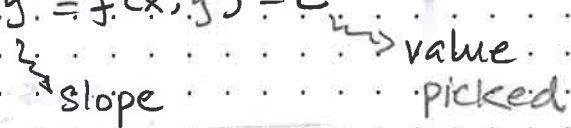
2. Compute $f(x, y)$ or slope

3. Draw line segments 

4. Draw lines tangent to the slope: Integral curve

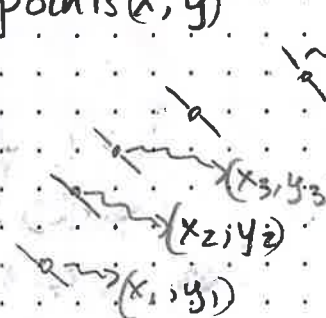

Particular solution of $y' = f(x, y)$

By human hand:

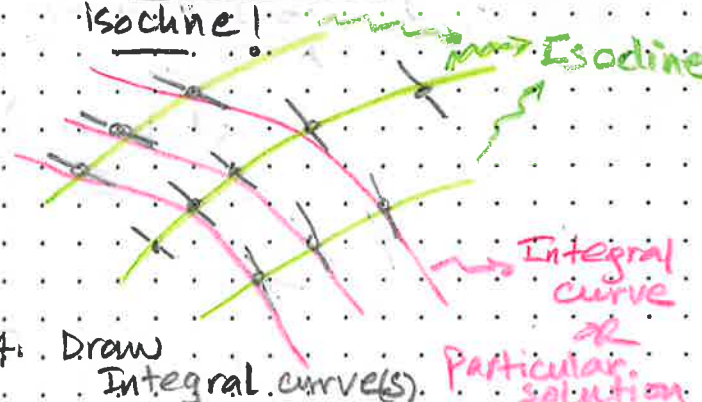
1. Pick a slope C , such that $y' = f(x, y) = C$


2. Find all points (x, y) where $f(x, y) = C$ by inputting x and calculating y , and

Draw line segments that represent slope at points (x, y)


 slope at each (x, y) is the same

3. Plot $f(x, y) = C$. It's an isochne!


 Isodine
 Integral curve or Particular solution

4. Draw Integral curve(s)

EXAMPLE ① $y' = -\frac{x}{y}$ → Solve this ODE graphically

Solution:

STEP 1. $y' = -\frac{x}{y} (= f(x,y))$

⇒ $-\frac{x}{y} = c$ → $f(x,y) = c$

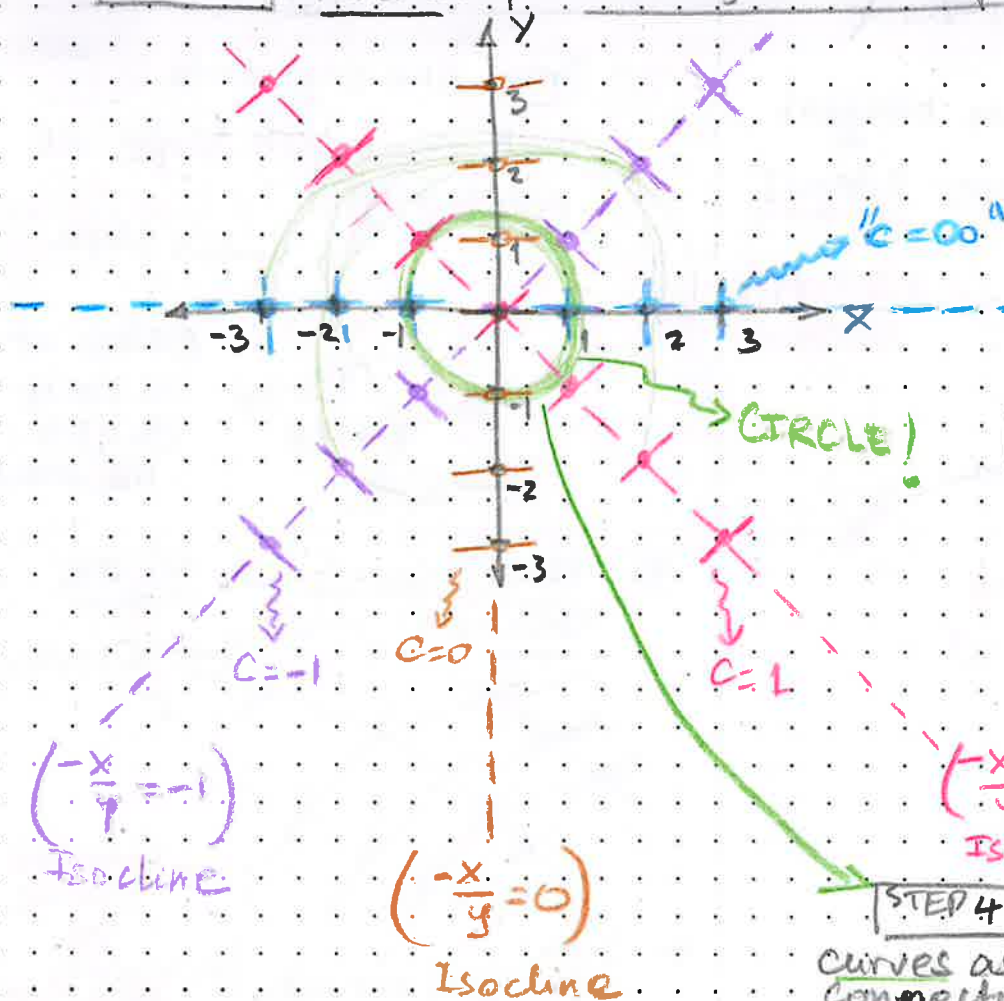
Pick value of slope, c : slope { looks like this is the equation of line: $y = -\frac{1}{c}x$ }

$c = [-1, 0, 1, \infty]$

$\{ \text{line segments parallel to } x\text{-axis} \}$
 $\{ \text{line segments parallel to } y\text{-axis} \}$

STEP 2. Find all points (x,y) with slopes $c = [-1, 0, 1, \infty]$

And draw line segments representing slope, c .



STEP 3. Plot Isoclines for each $f(x,y) = c$

STEP 4. Draw integral curves as a tangent lines connecting slopes.

Compare graphical solution to the analytical solution. (25) KVB

START: ODE $y' = -\frac{x}{y} \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$

$$\Rightarrow y dy = -x dx$$

Integrating on both sides

$$\Rightarrow \int y dy = \int -x dx + P \rightsquigarrow \text{arbitrary integration constant}$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + P$$

$$\Rightarrow x^2 + y^2 = 2P$$

END: $\Rightarrow x^2 + y^2 = (\sqrt{2P})^2 \rightsquigarrow \text{Equation of a circle!}$

EXAMPLE ② Solve the following ODE graphically:

$$y' = 1 + x - y$$

Solution:

STEP 1.

$$y' = 1 + x - y = f(x, y)$$

$\Rightarrow 1 + x - y = C$ slope of the ODE

$$\Rightarrow y = x + (1 - C)$$

Pick values for slope, C : $C = [-1, 0, 1, 2]$

Let's find equations corresponding to $C = [-1, 0, 1, 2]$:

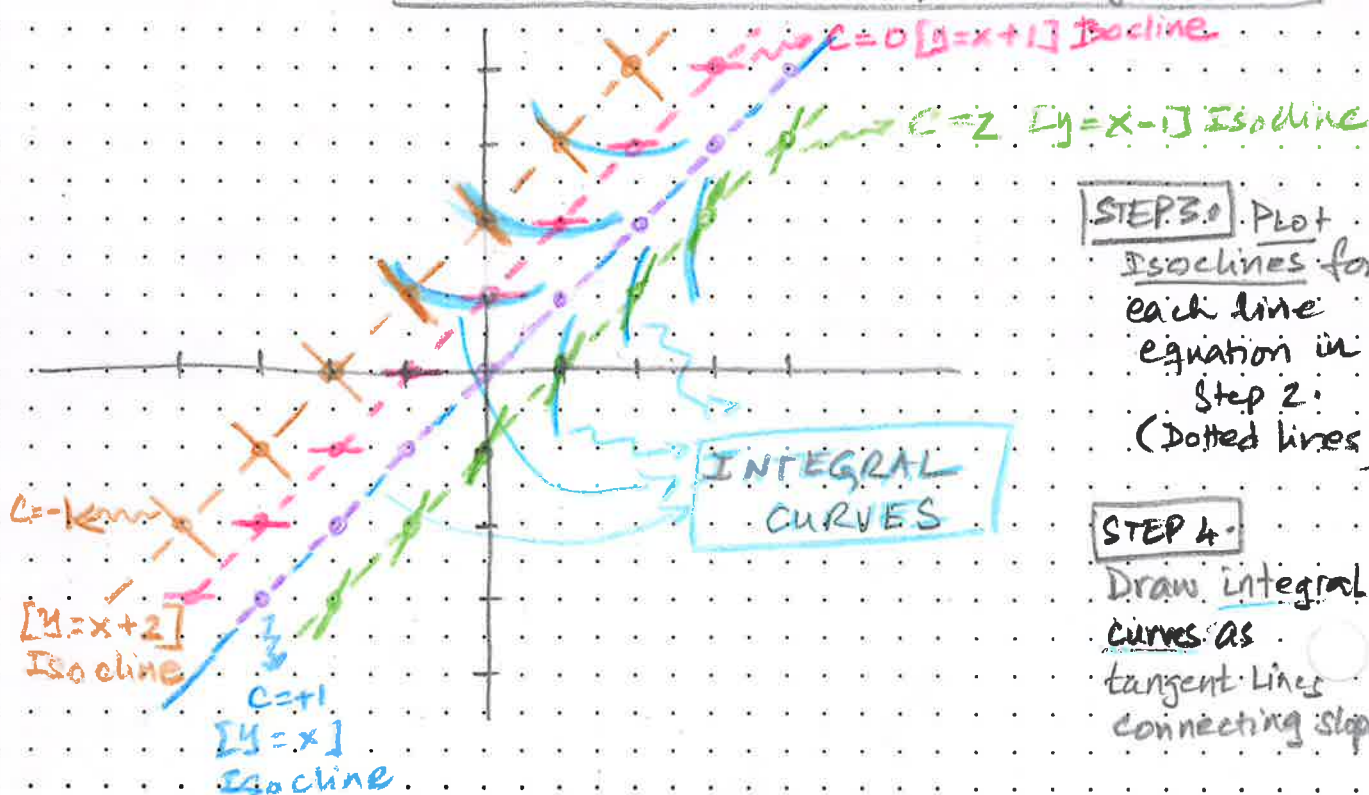
$$C = -1 \Rightarrow y = x + 2$$

$$C = 0 \Rightarrow y = x + 1$$

$$C = +1 \Rightarrow y = x$$

$$C = 2 \Rightarrow y = x - 1$$

STEP 2. Find all points (x, y) with slopes, $C = [-1, 0, 1, 2]$ and draw line segments representing, slope, C :

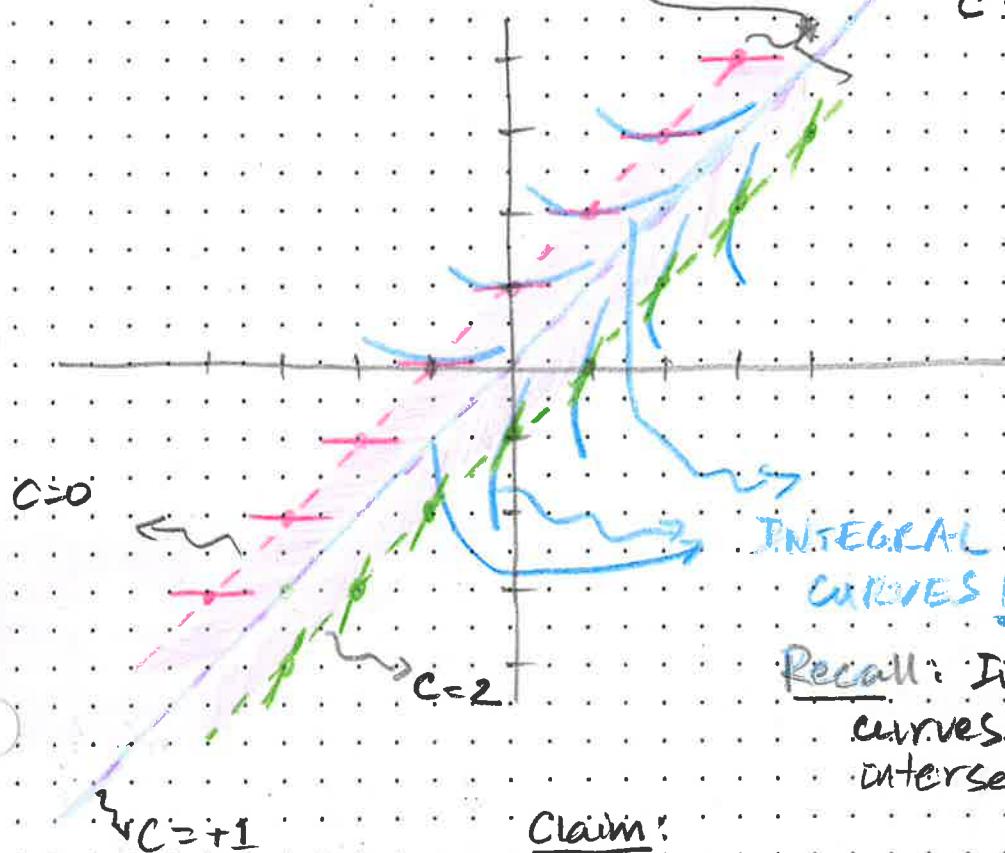


STEP 3. Plot Isoclines for each line equation in Step 2. (Dotted lines)

STEP 4. Draw Integral curves as tangent lines connecting slopes

Review the graphical solution:

→ There is a corridor between isoclines $C=0$ & $C=+2$



Recall: Integral curves cannot intersect!

Claim:

→ Isocline, $C=+1$ with equation $[y=x]$ is an integral curve.

Evaluate graphically generated integral curve $[y=x]$

using the analytical approach:

START: ODE: $y' = 1 + x - y$

→ Observe on graph, $y=x$ is an integral curve

→ Integral curve is a particular solution.

→ $y_1(x) = x$

Substitute into ODE $\Rightarrow y_1' = 1 + x - y_1$

$$\Rightarrow \frac{d}{dx} x = 1 + x - x \Rightarrow 1 = 1$$

END!

$y=x$ is an (accidentally discovered) particular solution of ODE: $y' = 1 + x - y$

